Mohammed Rishin Poolat

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Professional Summary

AI and Data Science Researcher working on wearable AI for precision health, building and training machine learning models on wearable sensor data and developing AI agent systems with LLMs, multi-LLM orchestration and the Model Context Protocol (MCP). Brings a strong engineering background in deploying LLM-powered agents and data pipelines to production, with a focus on models and tools that are reproducible, scalable and useful beyond a notebook. MSc in Advanced Computer Science from the University of Leicester, UK, and AWS Certified Cloud Practitioner.

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, C#, HTML/CSS
Machine Learning & Data Science: Machine Learning, Deep Learning, Time-Series Analysis, Wearable Sensor Data, NLP, Computer Vision, PyTorch, TensorFlow, scikit-learn, Keras, LightGBM, DeBERTa, NumPy, Pandas, OpenCV, Jupyter
Generative AI & Agents: LLMs, AI Agents, Multi-LLM Orchestration, MCP, RAG, Vector Databases, OpenAI API, Google Gemini API, Claude Code
Backend & Frontend: FastAPI, Node.js, Express.js, ASP.NET Core, React.js, Next.js, Tailwind CSS, Socket.IO
Databases & Cloud: PostgreSQL, MongoDB, MySQL, DynamoDB, Redis, Supabase, AWS (EC2, RDS, S3, Lambda, CloudFront)
DevOps & Tools: Docker, Git, CI/CD, Linux, RabbitMQ, BullMQ, n8n, Puppeteer

Experience

Weill Cornell Medicine-Qatar **November 2026 – Present**
Researcher, AI & Data Science *Doha, Qatar*

- Build and train machine learning models on wearable sensor data to support research in wearable AI for precision health.
- Process and analyse continuous, high-frequency physiological time-series data from wearable devices to prepare it for modelling and evaluation.
- Develop AI agent systems using LLMs, multi-LLM orchestration and the Model Context Protocol (MCP) to turn continuous health data into meaningful insights.

Acugence Systems **June 2026 – October 2026**
Software Engineer *Doha, Qatar*

- Led client discovery sessions to turn business needs into structured project plans, and built responsive employee-facing portals across UI/UX, front end and database models.
- Developed custom Python modules to extend Odoo ERP across HRMS, time off, payroll and maintenance, tailoring workflows, business logic and reporting to each client.
- Configured and standardized the team's Claude Code setup, establishing shared conventions and tooling so AI-assisted development could be adopted consistently.

OffSeason **March 2024 – December 2025**
Software Engineer *Leicester, United Kingdom*

- Engineered an AI-powered WhatsApp assistant using OpenAI models and the Model Context Protocol (MCP), letting the team query live sales, inventory and order data across multiple Supabase databases in plain English.
- Built an OpenAI-powered service that converts production errors into detailed GitHub issues, and automated order confirmations, payment-failure alerts and daily sales reports via WhatsApp and email using Trigger.dev, n8n and Resend.
- Created web-scraping and data-pipeline services with Puppeteer and Bun to collect product and pricing data from external sources through scheduled background jobs.

- Built a full-stack e-commerce platform (Next.js, TypeScript, Stripe) and Tracy, an in-house POS and inventory system serving two retail stores with barcode checkout, Typesense search and four UK VAT schemes.

Qburst Technologies **July 2023 – December 2023**
Software Engineer *Kerala, India*

- Developed a full-stack content management system for internal teams using React, Node.js and PostgreSQL, with REST APIs to create, organize and publish content.
- Built a social media platform with relational data in AWS RDS, chat messages in DynamoDB and media served through S3 and CloudFront, deployed and scaled on EC2 with load balancing.

Obraz **August 2021 – June 2023**
Software Engineer *Kerala, India*

- Trained and fine-tuned machine learning models for the company's computer vision and AI projects, working across the model development lifecycle from data to deployment.
- Supported dataset preparation and annotation to produce clean, well-labelled training data, maintaining data quality and consistency across projects.

Education

University of Leicester **January 2024 – July 2025**
Master of Science in Advanced Computer Science *Leicester, United Kingdom*

APJ Abdul Kalam Technological University **July 2019 – June 2023**
Bachelor of Technology in Computer Science and Engineering *Kerala, India*

Certifications

AWS Certified Cloud Practitioner **Link**
Amazon Web Services (AWS) *Issued: September 2024*

Projects

Automated Essay Scoring System | *Python, DeBERTa, LightGBM, FastAPI* GitHub

- Combined a DeBERTa transformer ensemble with LightGBM feature models and calibration to grade essays, achieving 0.8358 QWK on 24,728 essays, served via FastAPI with a Next.js dashboard and Gemini-generated feedback.

Real-time Helmet Detector | *Python, TensorFlow, OpenCV, Raspberry Pi* GitHub

- Trained a CNN to detect whether riders wear helmets and deployed it on a Raspberry Pi with a camera and speaker for real-time on-device monitoring and audio feedback.

Meeting Scheduler Agent | *Python, Gemini API, Gmail & Calendar APIs* GitHub

- Built an autonomous scheduling agent that classifies emails into intents, books, offers, cancels and reschedules meetings, and works around LLM time-corruption by substituting verified values in Python.

AI-Powered RFP Management System | *TypeScript, LLMs, Node.js, RabbitMQ, Redis* GitHub

- Built a procurement platform where an LLM converts requests into structured RFPs, extracts vendor pricing from emails and scores proposals, with RabbitMQ queuing and Redis caching.

Property Transaction Extraction System | *Next.js, PostgreSQL, Redis, BullMQ* GitHub

- Engineered a document pipeline that parses property transaction PDFs, translates and extracts structured fields through queued background jobs, with a searchable dashboard.

Expense Tracker with Smart Receipt OCR | *.NET 9, React, Tesseract.js* GitHub

- Implemented a Clean Architecture expense tracker combining client-side Tesseract.js OCR with server-side OCR.space to auto-populate receipts, with fallback when OCR fails.

Penply – Collaborative Study Platform | *React, Node.js, Socket.IO, AWS* Live

- Created a study platform where students join rooms to chat in real time, share notes and files, and collaborate live, with Google OAuth and two-factor authentication.